

ComplaintSense: Consumer Complaint Classification — Stakeholder Engagement Plan

1. Selected Stakeholder Groups

- **Complainants (End Users):** The individuals submitting complaints in their own words. Their phrasing, category assumptions, and tolerance for misrouting directly shape what "good" classification means in practice.
- **Complaint-Handling Staff:** The people who currently triage complaints manually and would work alongside (not be replaced by) the classifier's routing output.

2. Stakeholder Priorities

Complainants care most about being **understood correctly and quickly**, particularly for urgent categories like `fraud_unauthorized` or `billing`. They are unlikely to phrase issues the way a training dataset expects — complaints often combine multiple issues in one message, use informal language, or omit context a model assumes is present. Their priority is that the system doesn't penalize them for writing naturally.

Complaint-handling staff care about **reducing repetitive manual sorting without losing control** over ambiguous or sensitive cases. They need to trust that a model's category boundaries (e.g. `warranty_repair` vs. `product_defect`, or `subscription_cancel` vs. `general_inquiry`) match how complaints are actually handled operationally, not just how they were labeled in the training data.

3. Potential Harms and Benefits

Complainants

- *Potential Harms:* Misclassification of complaints that mix multiple issues, or that use phrasing underrepresented in the small (380-row) training set, could delay resolution — especially damaging for time-sensitive categories like fraud or refunds.
- *Potential Benefits:* Faster, more consistent routing means less time spent navigating rigid category menus or repeating an issue to multiple support agents.

Complaint-Handling Staff

- *Potential Harms:* Given the model's current validation performance (Macro F1 = 0.2738), staff could inherit a meaningful volume of misrouted complaints, increasing rework rather than reducing it if the model is deployed without safeguards.
- *Potential Benefits:* Even an imperfect classifier can act as a first-pass sort, freeing staff time for judgment-heavy cases rather than routine categorization, provided low-confidence predictions are flagged for human review rather than auto-routed.

4. Selected Engagement Methods

- **Complainants:** Focus Groups, used to collect authentic, unedited complaint phrasing and surface where the current 10-category taxonomy fails to match how people actually describe problems.
- **Complaint-Handling Staff:** User Panel, run as a recurring review body rather than a one-off session, evaluating per-category model performance against real operational friction.

5. Method Justification

Focus Groups for Complainants The training data is small (380 labeled examples) and was not collected directly from this project's target users, which is a plausible contributor to the model's weak validation score. A structured focus group — rather than a passive survey — lets complainants react to real examples: shown an anonymized set of ambiguous or misclassified complaints from validation, they can explain in their own words what category they'd expect and why. This directly targets the dataset's biggest limitation: it doesn't capture how our actual user base phrases multi-issue or informally-worded complaints. Session transcripts become a source of new, realistic training examples rather than synthetic ones, and expose which categories are conceptually confusing to users (not just statistically hard for the model).

User Panel for Complaint-Handling Staff A single feedback session is not enough for people who would use this system daily. A standing user panel lets staff review the model's actual per-class precision/recall — for example, whether `general_inquiry` is disproportionately absorbing complaints that belong elsewhere — and connect that directly to what causes rework on their end. Because the current baseline is a classical ensemble (not the originally attempted transformer, which collapsed to predicting a single class), staff input is especially important for deciding a practical confidence threshold: below what score should a prediction route to a human instead of being trusted outright. This keeps the model positioned as a triage aid, not an unsupervised decision-maker.

6. Influence on Dataset and Model Design

Focus group transcripts will be used to expand and rebalance the training set, particularly for categories where participants' natural phrasing diverged most from the labeled examples — this may also prompt re-examining whether some of the 10 categories should be merged or split. User panel findings will directly inform a confidence-threshold policy for routing (low-confidence predictions sent to manual review rather than auto-assigned), and may surface mislabeled or ambiguous training examples that staff recognize as edge cases from real operations. Any complaint text or phrasing volunteered by participants during these sessions will be anonymized and only incorporated into the dataset with explicit consent, documented before use in training.